

Bank Client Segmentation

K-Medoids Clustering with Gower Distance



**TRANSFORMING CUSTOMER DATA INTO STRATEGIC INSIGHTS FOR
PERSONALIZED BANKING**

Dataset & Methodology

- *5,000 bank clients*
- *17 features*
- *12 normalized features*
- *Categorical : Gender, Job, Area, CitySize, Investments*



EDA



Preprocessing and Gower distance



Metrics evaluation



K-Medoids



Defining Personas & Business plan

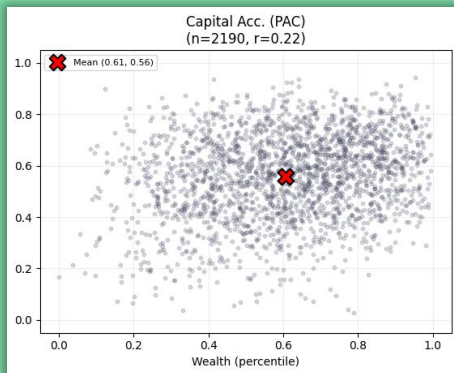
EDA: 3 Key Insights

Patterns that should emerge in the final clusters

01

PAC = Discipline, Not Wealth

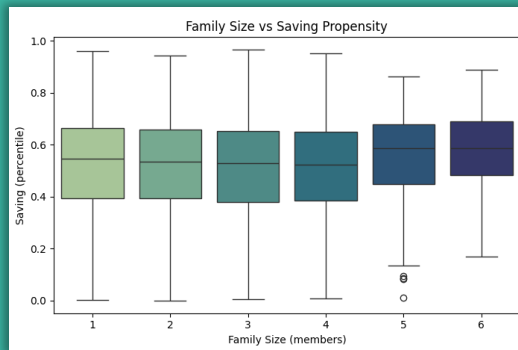
Lowest Wealth-Saving correlation ($r=0.22$).
PAC attracts savers at ALL wealth levels.



02

Precautionary Saving

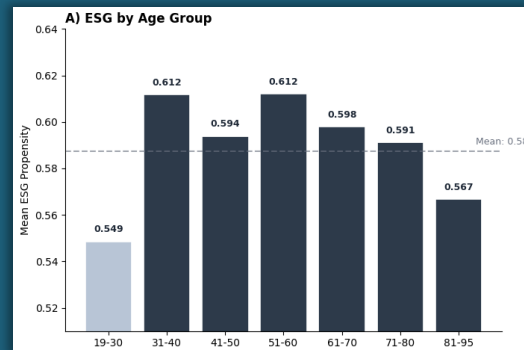
Larger families: MORE debt AND MORE
saving. Responsibility drives prudence.



03

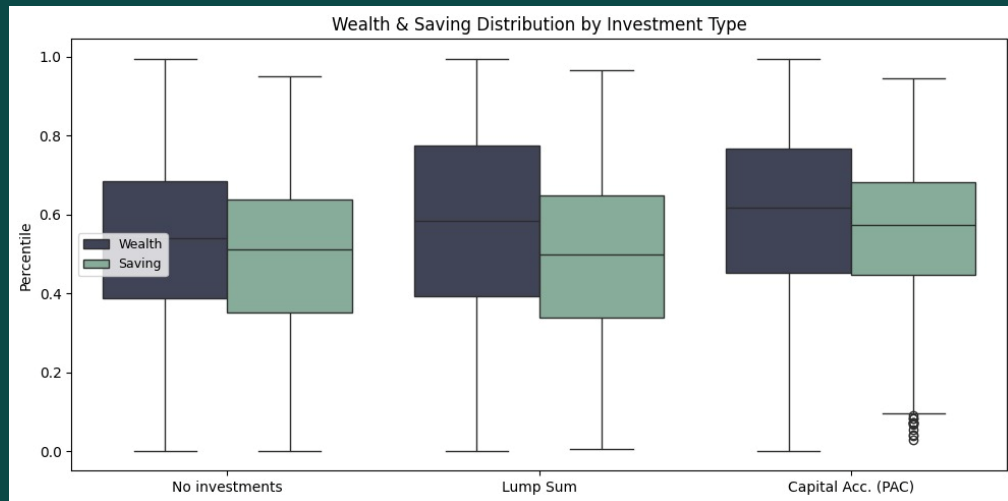
ESG Distribution

Young have LOWEST ESG because they
invest less.

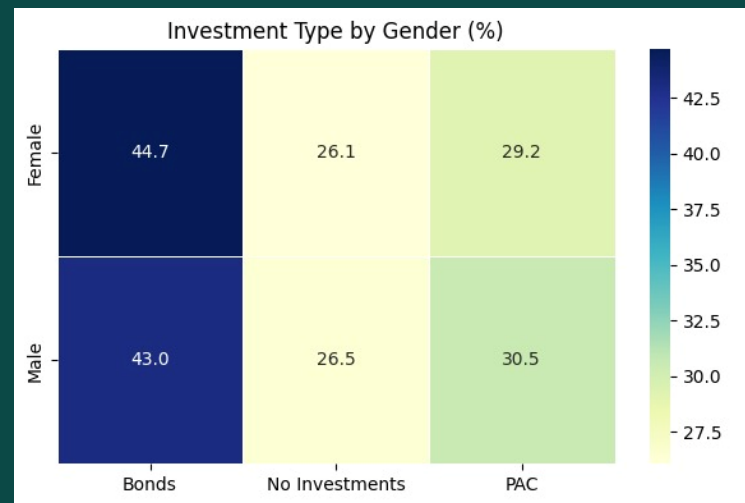


Weights deductions on Data

FINANCIAL FEATURES



DEMOGRAPHIC FEATURES



Signal of different groups crucial for clusterization

The 2 groups invest in the same fashion, making the genders irrelevant

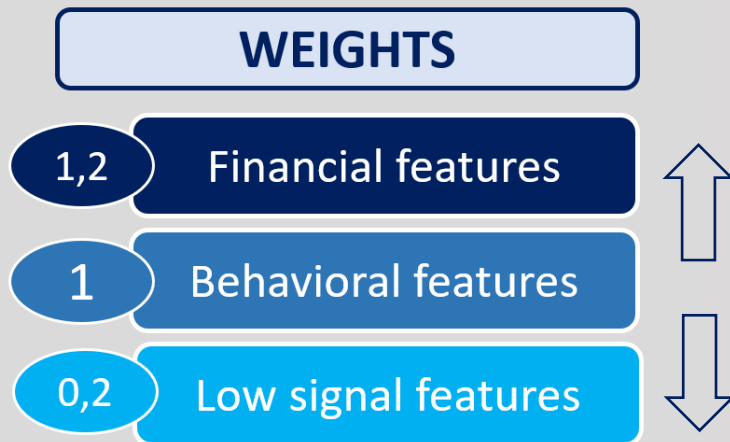
Preprocessing & Gower Distance

Gower Distance

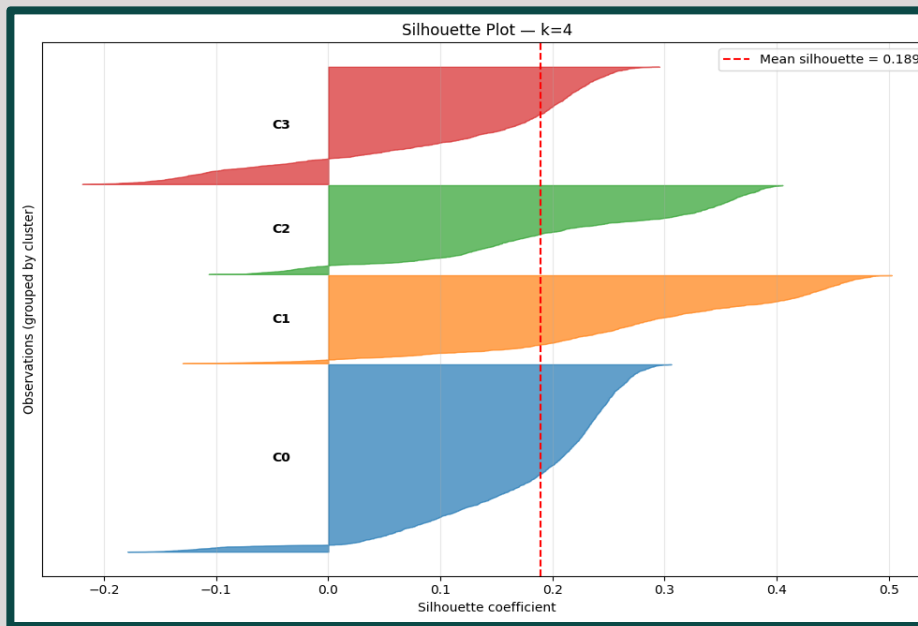
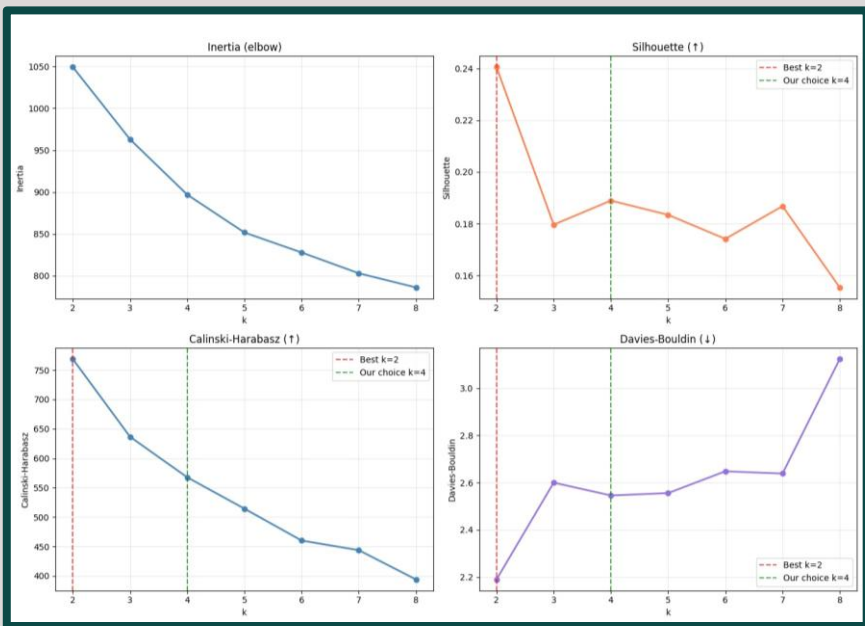
Purpose: Similarity metric for mixed data

Output: Matrix of normalized distances

RESULTS :
Min : 0,0213
Max : 0,7647
Mean : 0,3521

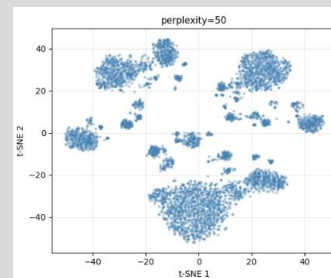
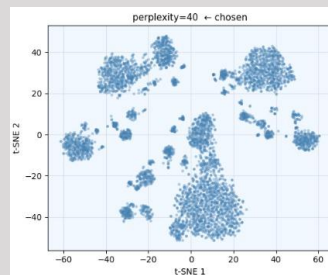
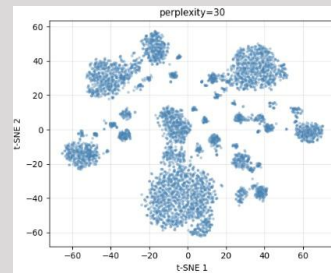
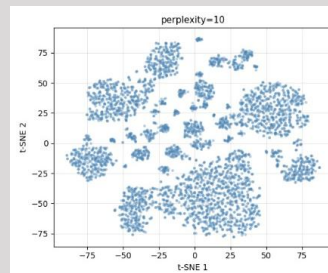
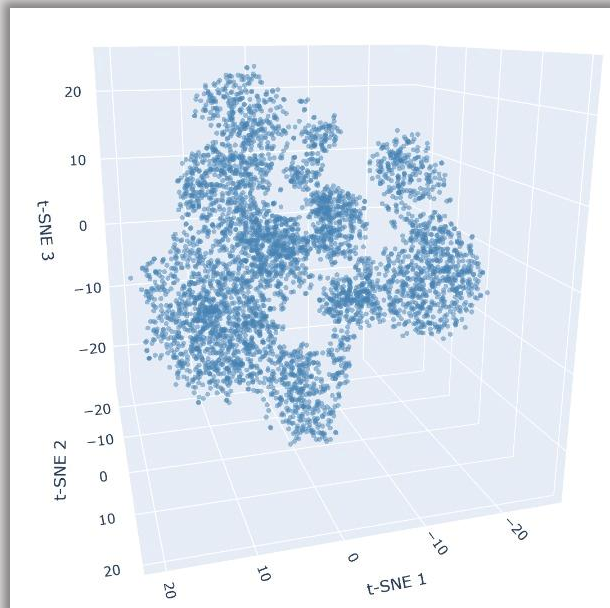


Metric evaluation



T-SNE 3D DATA PLOT

Non-linear dimensional reduction with precomputed Gower distance



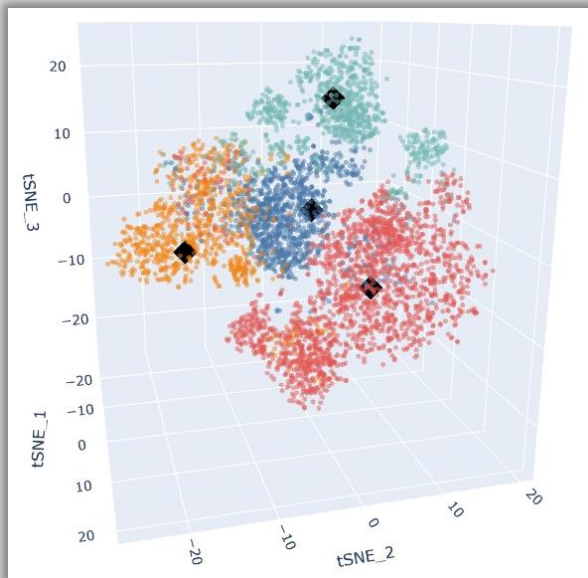
PERPLEXITY = 40

METRIC:
'Precomputed'

RANDOM STATE = 42

K – MEDOIDS: RESULTS

MEDOIDS are the REAL client of the DataSet



CLUSTER SIZES

0 1,681 (33,6%)

1 1,257 (25,1%)

2 865 (17,3%)

3 1,197 (23,9%)

Algorithm:
K-Medoids (PAM)

Distance:
Gower

Number of clusters:
k = 4

Silhouette:
0,1463

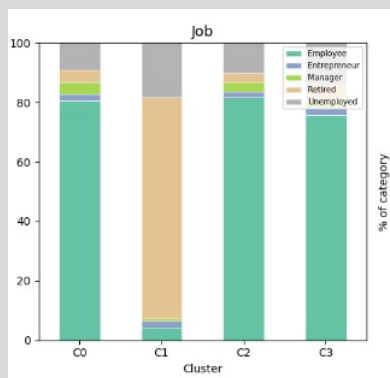
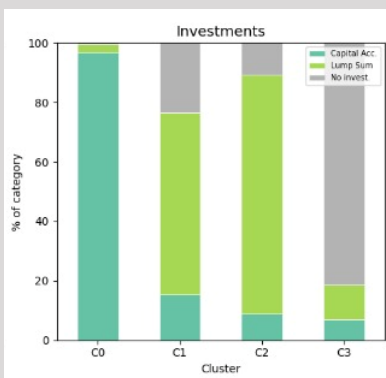
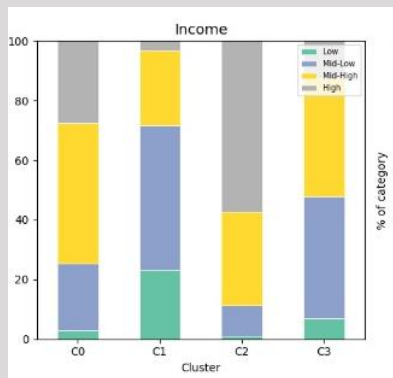
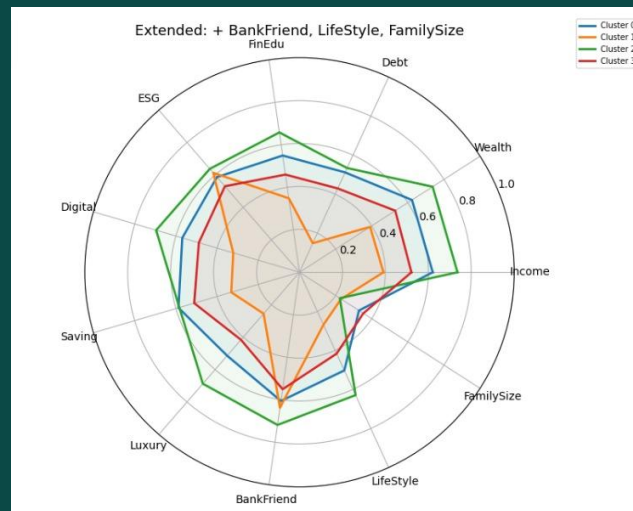
Cluster profiling

The extremes:

- **Elderly conservatives**
- **Young Affluent**

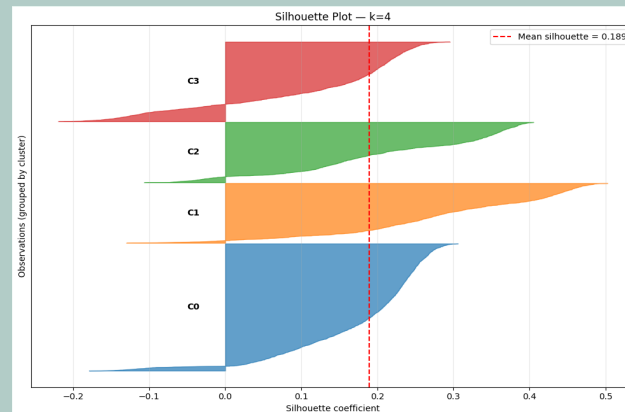
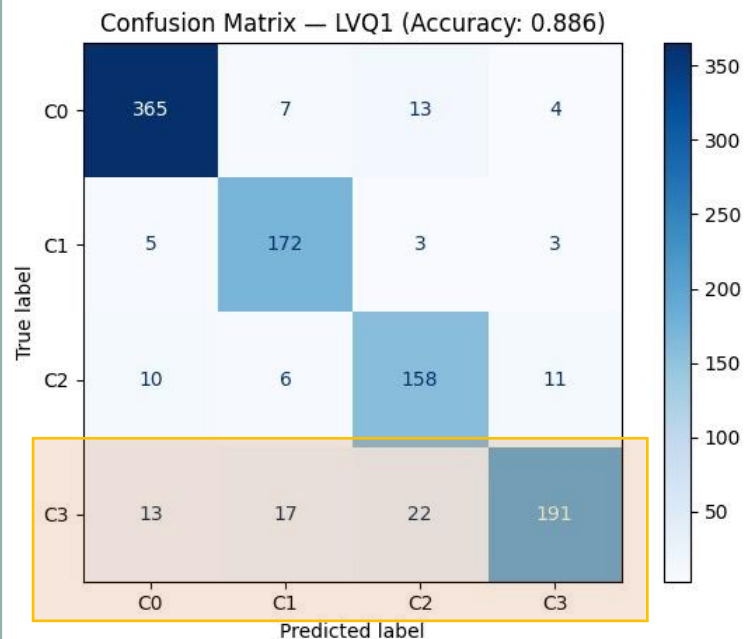
The middle:

- **Mass active**
- **Liquidity Holders**



	Gen_L	Job_L	Inv_L	Inv_%	Age
Cluster					
0	M	Employee	CAP	96.8	56
1	F	Retired	Lump sum	61.3	80
2	F	Employee	Lump sum	80.2	53
3	F	Employee	No inv.	81.5	58

LEARNING VECTOR QUANTIZATION (LVQ)



	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Age	-4.146	-10.825	-3.431	0.563
FamilySize	0.657	0.088	1.106	0.880
Income	-0.005	-0.013	0.029	0.073
Wealth	-0.187	-0.014	-0.030	0.020
Debt	-0.047	-0.083	0.044	0.008
FinEdu	0.076	-0.118	0.063	0.034
ESG	0.014	-0.019	-0.028	-0.058
Digital	-0.033	-0.133	-0.015	0.009
BankFriend	0.079	-0.076	-0.092	0.094
LifeStyle	-0.043	-0.044	-0.069	0.140
Luxury	0.005	-0.047	0.032	0.001
Saving	-0.042	-0.028	0.051	0.114

The 4 Personas

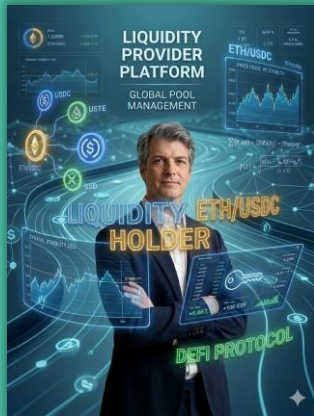


The Elderly Conservatives

Age: 80y | Lump Sum | Debt : 0.15

Behaviour :

They are in a capital preservation phase



Liquidity Holders

Age: 58y | No investment | FinEdu: 0.46

Behavior:

Unengaged clients with no investments



Young Affluent Investors

Age: 52y | Lump Sum | Income: 0.74

Behavior:

Highly active and growth oriented



Systematic Savers

Age: 57y | PAC | Savings: 0.59

Behavior:

Systematic investments via PAC

Personalized banking proposal

**Elderly
Conservatives**

Dedicated physical advisory. Capital-guaranteed products such as green bonds.

Liquidity Holder

Low friction entry products such as micro-PAC.

**Young Affluent
Investors**

Active Wealth Management, niche high potential alternatives.

Systematic Savers

Loyalty rewards and complementary pension funds.

Thank you!

Simone Cerrone – Stefano De Amici – Riccardo Cataldi – Francesco Villani – Vittoria Tomasini



Business Case 1 | ML for Fintech | Politecnico di Milano